

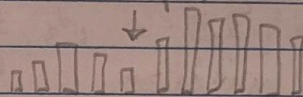
Binary Images

Can only have two values (0 or 1), simple to process

Binary Image $b(x,y)$ is usually obtained from Gray-level image $g(x,y)$ by Thresholding

$$\text{e.g. } b(x,y) = \begin{cases} 0 & g(x,y) < T \\ 1 & g(x,y) \geq T \end{cases}$$

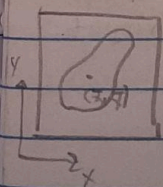
Compute histogram (# of pixels vs gray level)
- pick in valley



But simple thresholding doesn't always work. To help, use backlighting

$I=b(x,y)$ Geometric Properties

Assume $b(x,y)$ is continuous w/ only one object



Area (0th moment) $A = \iint_I b(x,y) dx dy$

Position: center of area (first moment)

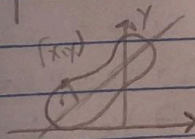
$$\bar{x} = \frac{1}{A} \iint_I x b(x,y) dx dy \quad \bar{y} = \frac{1}{A} \iint_I y b(x,y) dx dy$$

Orientation: difficult to define!

- use axis of least second moment
(minimum effort to spin around)

- minimizes

$$E = \iint_I r^2 b(x,y) dx dy$$



what to
use for
axis

$y = mx + b$? no bc $m \in (-\infty, \infty)$

use: $x \sin \theta - y \cos \theta + p = 0$ where p, θ are finite

Goal: Find p and θ that minimize E

$$r = \frac{|ax+by+c|}{\sqrt{a^2+b^2}} \text{ given line } ax+by+c=0$$

Similarly, given $x \sin \theta - y \cos \theta + p = 0$

$$r = \frac{|x \sin \theta - y \cos \theta + p|}{\sqrt{\sin^2 \theta + \cos^2 \theta}} = |x \sin \theta - y \cos \theta + p|$$

So minimize

$$E = \iint (x \sin \theta - y \cos \theta + p)^2 b(x, y) dx dy$$

Using $\frac{\partial E}{\partial p} = 0$ we get $A(x \sin \theta - y \cos \theta + p) = 0$
where $(\bar{x}, \bar{y})$ is COM $\rightarrow$ axis passes through it

Change coordinates:

$$x' = x - \bar{x} \quad y' = y - \bar{y}$$

$$x \sin \theta - y \cos \theta + p = x' \sin \theta - y' \cos \theta$$

So $E = a \sin^2 \theta - b \sin \theta \cos \theta + c \cos^2 \theta$

$$a = \iint (x')^2 b(x, y) dx' dy'$$

$$b = 2 \iint x' y' b(x, y) dx' dy'$$

$$c = \iint (y')^2 b(x, y) dx' dy'$$

$$\frac{dE}{d\theta} = (a-c) \sin 2\theta - b \cos 2\theta = 0 \rightarrow \tan 2\theta = \frac{b}{a-c}$$

θ has 2 solns! (1-max, other-min)

$$\frac{dE}{d\theta} = (a-c) \cos 2\theta + b \sin 2\theta$$

Discrete Binary Images

b_{ij} = value of pixel at row i , col j

$$\text{COM: } \bar{x} = \frac{1}{A} \sum_{i=1}^m \sum_{j=1}^n i b_{ij} \quad \bar{y} = \frac{1}{A} \sum_{i=1}^m \sum_{j=1}^n j b_{ij}$$

second moments

$$a' = \sum_{i=1}^m \sum_{j=1}^n i^2 b_{ij} \quad b' = 2 \sum_{i=1}^m \sum_{j=1}^n i j b_{ij} \quad c' = \sum_{i=1}^m \sum_{j=1}^n j^2 b_{ij}$$

Segmentation

A and B are connected if $\exists$ a path from A to B along which $b(x, y)$ is constant

Region Growing Algorithm

1. Find "unlabeled" seed point w/ $b=1$.
If not found, terminate
2. Assign new label to seed point
3. Assign same label to its neighbors w/ $b=1$
4. Assign ^{same} label to neighbors of neighbors w/ $b=1$. Repeat until none left
5. Go back to A

What do we mean by neighbor?

-4-connectedness 

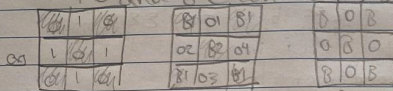
-8-connectedness 

-neither is perfect

Jordan's Curve Theorem

-a closed curve creates 2 connected regions

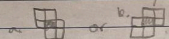
-4-C and 8-C contradict this



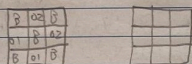
img

why bad (not loop separate) (background is connected)

-introduce asymmetry w/ 6-C



now:

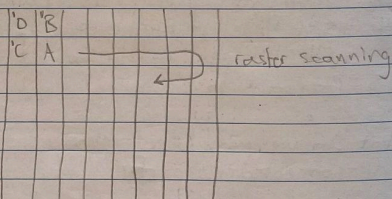


6-Ca

6-Cb

-makes square grid like hexagonal grid

Sequential Labelling



Look just at already labeled pixels

If $A=0 \Rightarrow \text{label}(A) = \text{background}$

$\Rightarrow \text{label}(A) = \text{new label}$

$\Rightarrow \text{label}(A) = \text{label}(D)$

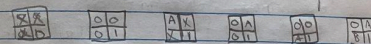
$\Rightarrow \text{label}(A) = \text{label}(C)$

$\Rightarrow \text{label}(A) = \text{label}(B)$

if $\text{label}(A) = \text{label}(C) \Rightarrow \text{label}(A) = \text{label}(C)$

-what if $\text{label}(A) \neq \text{label}(C)$? $\rightarrow$ equivalence table

note $\text{label}(A) = \text{label}(C)$ and assign $\text{label}(A) = \text{label}(C)$



1	[1]	0*	1	-		my val to
2	[2]		2	-		its index
3	[1]	s=3	3	1	1	close val
4	5	s=1	4	-	[2]	
5	2		5	2,4	[2]	
6	3		6	3	[1]	
7	4		7	4	[2]	
8	9		8	-	[2]	
9	7		9	7,8	[2]	
10	6		10	6	[1]	
11	8		11	8	[2]	
12	11		12	11	[2]	